

Project Development Phase

Preprocessing Steps & Business Questions

Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites (2019)

Project Title	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites (2019)
Domain	Culture & Heritage Analytics / Tourism & Conservation Business Intelligence
Tool	Tableau Desktop / Tableau Public
Dataset	Kaggle UNESCO World Heritage Sites Dataset, 1978–2019 (~1,121 sites, 12 fields)
Prepared By	M. Deepika (Team Lead), M. Supraja, K. Srikala, I. Pushpa, U. Bhavana
Mentor	Not yet assigned
GitHub Repository	https://github.com/mankudeepika123-rgb/Heritage-Treasures-An-In-Depth-Analysis-of-UNESCO-World-Heritage-Sites-2019-
Tableau Public Link	https://public.tableau.com/app/profile/manku.deepika5282/viz/HeritageTreasuresAnIn-DepthAnalysisofUNESCOWorldHeritageSites2019/UNESCOSiteOverviewDangerSitesAnalysis
Date	22/08/2026

5.1 Data Preprocessing Steps

The following preprocessing steps were carried out before any chart was built, to ensure the UNESCO dataset was clean and correctly modeled in Tableau.

- Reviewed the site-level dataset (~1,121 rows, 12 fields) for missing values, duplicate site names, and inconsistent Country/Region naming
- Standardised Country and Region names to match Tableau's built-in geographic roles
- Set the geographic role of the Country field to Country/Region so the World Map renders without unmatched locations
- Converted Year Inscribed to a Date/Year field to enable time-based filtering
- Created a calculated field for Danger Status flag (In Danger / Not In Danger) from the raw danger indicator column
- Created a Years-Since-Inscription calculated field to support the inscription trend analysis
- Verified Category values (Cultural, Natural, Mixed) matched exactly across all sheets before building shared filters

5.1.1 Business Questions

The preprocessing steps above were driven by the following business questions, each of which maps to a chart, filter, or dashboard defined in the Project Design Phase.

#	Business Question	Answered By
BQ-01	Which regions and countries carry the largest share of sites currently on the Danger List?	World map by Danger Status + Region-wise Danger ranking chart
BQ-02	How are World Heritage Sites distributed across the Cultural, Natural, and Mixed categories?	Category distribution chart
BQ-03	How has the rate of new site inscriptions changed from 1978 to 2019?	Inscription trend line chart

#	Business Question	Answered By
BQ-04	Which countries hold the most inscribed sites, and how does that compare to their danger-listed share?	World map filterable by Country + Region-wise Danger ranking chart
BQ-05	Is there a relationship between how long ago a site was inscribed and its likelihood of being danger-listed?	Inscription trend chart cross-referenced with Danger Status filter
BQ-06	Which stakeholder-specific view best supports funding, tourism, and policy decisions respectively?	Conservation Officer, Tourism Planner, and Policy Researcher dashboards